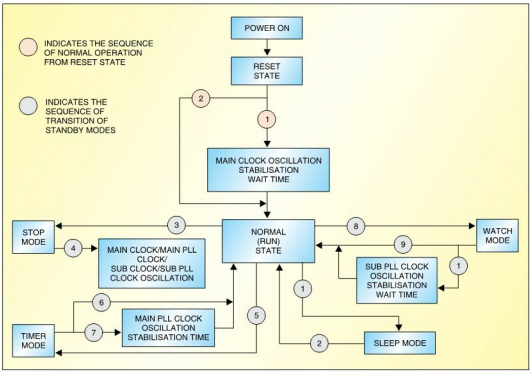


Fig. 5: Sequence of mode-state transitions

With interrupts, the CPU can sleep all the time and wake up only when an interrupt occurs.

Peripherals. Active peripheral power consumption should be considered when choosing a low-power MCU, as it is one of the biggest contributors to sleep-mode current budget. Some of the low-power MCUs are retrofitted versions of older architectures that were not designed for low power. The ability to individually enable or disable the peripherals when needed, or more importantly atomic enabling or disabling of the peripherals, is a key feature. Enabling internal analogue or digital modules can cause a significant increase in the overall current consumption. However, it can be difficult to evaluate. Well-documented MCUs describe this additional current consumption in their datasheet, while others claim this to be zero, which is very seldom a correct statement and the MCU is likely to have poor performance. The power consumption in analogue modules is static.

There is often a trade-off between power consumption and robustness, accuracy, speed and fast start-up time in analogue modules. Decreasing power consumption often decreases the quality of the analogue module. On the other hand, power consumption in digital logic is mainly due to the toggling frequency, capacitive load and supply voltage. Serial interface logic can allow shifting out of data in background, while the MCU is powered down.

Another key benefit that peripherals can provide is interoperability. For instance, the MSP430 ADC12 module can be triggered either by timer A or timer B.

Peripheral clock control is another feature that reduces current consumption of the system.

Brownout protection. The single-most important analogue module in terms of power consumption during sleep mode is the BOD. Many MCUs have built-in BOD, which resets the MCU when the power supply dips below the normal operating range. This keeps the MCU in a defined state when the Vcc is below its operating threshold.

The BOD is not important to the MCU while it's in sleep mode but it is extremely important when it wakes up. Therefore most MCUs keep the BOD active during sleep mode because brownout cannot be predicted and it contributes substantially to sleep-mode power consumption. Mostly, the BOD circuit requires an additional 10-70 µA of current for operation. This adds the brownout current consumption directly to the bottom line of the power budget of the system.

There are two ways of getting around BOD power consumption in sleep mode: making a 'zero power' BOD or turning the BOD 'on' and 'off.' Zero-power BOD may seem like the most attractive option, but lowering the power to the analogue module can make it very slow to respond quickly to an out-of-range voltage supply. On other hand, disabling the BOD when it enters sleep mode and enabling it again just before leaving sleep mode is a better approach as it ensures that the BOD is functioning when it is needed without causing any current penalty.

Pin leakage. Leakage current is sometimes overlooked when choosing a low-power MCU, but it must be considered for demanding low-power applications. Temperature, supply voltage and process technology affect the leakage current and result in increased sleep current. Most low-power MCUs have specified input leakage currents of 1 µA and could consume up to NµA for an N-input device.

Processing efficiency. Processing efficiency of the MCU is often misunderstood. It is often thought that 16-bit